

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element H Initial Template

Element H: Prototype Testing and Data Collection Plan

	Element C High Priority requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
1	Must be durable	For pvc element: 1) Grab a hold of one end of the prototype 2) Think about a situation that made you mad 3) Hit everything in a nearby vicinity as hard as you can until your anger subsides For aluminum wire element: 1) Have 2, ore more, people each firmly grabbing the hooks on either side of the wire 2) Have each person pull away from the other side as hard as they can, twisting and flexing the wire in the process	5 trials, no test cases	For PVC: - No dents or cracks - In one piece For aluminum wire: - Bending but no breaking - The wire segments will not separate	Fail: one or more parts of the prototype breaks or is damaged to the point of non-function Pass: All parts remain intact and are useable enough/damages are easy to repair Goal: all parts of the prototype remain unchanged
2	Moldless	For pvc element: 1) Use the prototype and after a few days of use, test to see if mold built up.	Trial 1: Pipe is angled at 45° Trial 2: Pipe is angled at 20° Trial 3: Pip is angled at 10°	For PVC: - No mold - Water fully empties - out after usage	Fail: the prototype develops mold during testing Pass: there is no evidence of mold growth anywhere on the prototype

					Goal: the prototype remains moldless after testing is over
3	Repairable	For pvc element: 1) Deconstruct the pvc pipe and attempt to reconstruct it 2) Attempt to remove and replace each part on Pvc pipe For aluminium wire element: 1) Attempt to remove the construction from it's hangers 2) Remove the hangers and replace them 3) Replace the construction to it's original place	3 trials, no test cases	For PVC pipe: All parts are able to be replaced without changing the functionality of the design. All parts can be easily deconstructed and replaced For aluminium wire: Can be removed from its position in the greenhouse, replaced, and not bend while doing so.	Fail: Any of the parts cannot be easily removed or replaced from the construction, or if deconstructing and reconstructing the design causes damage to any parts or limits their functionality. Pass: All parts on the construction and wire hangers remain unchanged Goal: The prototype can be deconstructed and reconstructed easily

Testing Details (H1, H2, H3, H4)

Feedback From Field Experts (H5)

Name of your field expert:

Brother

Dad

Why are they a field expert?

He likes to grow food, including tomatoes, which are grown by the club.

He is in STEM/engineering

What did they say to you about the tests you are proposing?

"Why is there a durability test, it's a water sprinkler, it's just gonna sit there, it's not a sword."

"Looks good"

Revised Testing Details That Incorporate Field Expert Feedback (H5)

	Element C High Priority requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
1	Must be easy/intuitive to use	1) Find a person 2) Explain that the prototype waters plants(do not explain how to use it) 3) Ask them how they think it should be installed/how they would use it	6 trials, 2 greenhouse club members 1 Mr. Doffermyre 3 random people who have no affiliation with the above groups or any gardening experience	The club members and Mr. Doffermyre will know how to use it easily Random people may have some trouble understanding it	Fail: At least 3 people or any of the club/Mr. Doffermyre do not know how to use it intuitively Pass: A majority of people, including the club and Mr. Doffermyre are able to figure it out easily

					Goal: all subjects are able assume how to use it quickly and correctly
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Element H: Prototype Testing and Data Collection Plan						
	5	4	3	2	1	0
1. Addressing Requirements. The final prototype Testing Plan addresses _____ of the <i>high priority</i> design requirements.	<i>all or nearly all</i>	<i>many</i>	<i>some</i>	<i>a few</i>	<i>one</i>	<i>none - or is missing altogether</i>
2. Conduct of Tests. The Testing Plan _____ in describing the conduct of tests	<i>is effective</i>	<i>is generally effective</i>	<i>is only adequate</i>	<i>is partially effective</i>	<i>is at least minimally effective</i>	<i>is void</i>

(through physical and/or mathematical modeling).						
3. Quantity of Tests Planned. The Testing Plan defines _____ of <i>all</i> design requirements based on the instructional context.	sufficient quantity to feasibly test 100%	sufficient quantity to feasibly test 80%	sufficient quantity to feasibly test 60%	sufficient quantity to feasibly test 40%	<i>sufficient quantity to feasibly test 20%</i>	<i>no feasible test</i>
4. Logic of the Plan. The Testing Plan provides _____.	a logical and <i>well-developed</i> explanation	<i>a logical and generally developed</i> explanation	<i>a generally logical and adequately developed</i> explanation	<i>a somewhat logical and partially-developed</i> explanation	<i>at least a generally logical and/or partially-developed</i> explanation	<i>no explanation</i>
5. Use of Field Experts. The Testing Plan is confirmed by _____.	<i>more than two</i> field experts with feedback provided and incorporated	<i>two</i> field experts with feedback provided and incorporated	one field expert with feedback provided <i>and incorporated</i>	one field expert <i>with feedback provided</i>	<i>one field expert but no response was documented</i>	<i>Field expertise is not mentioned or addressed and field experts are not consulted.</i>